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Understanding the interplay of Zn vapours and formation of intermetallic compounds during remote laser welding of Zn-coated steel to aluminium in a zero-gap lap joint configuration

A. Baghbani Barenji^{a,*}, P. Franciosa^a, S. Jabar^a, H. R. Kotadia^b, D. Ceglarek^a, K. F. Ayarkwa^c and J. R. Smith^c

^aWMG, The University of Warwick, Coventry, CV4 7AL, UK

^bSchool of Engineering, Liverpool John Moores University, Liverpool, L3 3AF, UK

^cJaguar Land Rover Limited, Abbey Road, Whitley, Coventry, CV3 4LF, UK

* Corresponding author. E-mail address: Ali.Baghbani-Barenji@warwick.ac.uk

Abstract

Remote laser welding (RLW) of Zn-coated steel to aluminium is challenged by: (1) Zn vaporisation causing spatters and poor seam surface when welding in a zero-gap lap joint configuration; and (2) brittle intermetallic compounds (IMCs) at the weld interface, leading to cracks and poor weld integrity. This paper aims to study the impact of Zn coating during the RLW of Zn-coated steel to aluminium in a nominal zero part-to-part gap lap configuration. The combination of metallurgical and mechanical analysis with an in-process monitoring approach, using a high-speed camera, allowed the study of the interplay between Zn formation/out-gassing, molten pool dynamics and formation of IMCs. Findings revealed that the presence of a Zn coating results in an un-even connection area between steel and aluminium, with localised lack of fusion. A significant 72% joint efficiency compared to parent aluminium was achieved when Zn is removed, opposed to 65% when Zn is present.

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Keywords: remote laser welding; Zn-coated steel to aluminium; intermetallic compounds; laser beam oscillation; zero-gap lap joint configuration; weld integrity.

1. Introduction

In the pursuit of carbon neutrality, manufacturing processes face a critical mandate to maximise product performances while reducing cost, system complexity and energy consumption. This is particularly important in the manufacturing of electric vehicles that require lightweighting and optimised material efficiency via dissimilar material structures to produce functional parts (tab connectors, enclosures, etc.). By complementing the properties of the individual materials and allowing optimisation of functional performances, joining of dissimilar materials represents a strategic area of innovation. Joining of steel to aluminium has been investigated for many years and attracted significant interest in research and industry for welding the battery packs

and automotive body structures [1]. The current state-of-the-art for joining these materials is mechanical joining because it avoids liquid phase reactions. Nonetheless, mechanical joining is not recommended for joints that must be airtight/watertight. Bearing benefits and drawbacks, other solutions have been studied, including chemical bonding (e.g., adhesive bonding), bimetallic inserts, solid-state welding (e.g., friction stir welding), and fusion welding (e.g., laser beam welding, cold metal transfer (CMT) [2].

Within fusion welding, remote laser welding (RLW) offers lower heat input compared to others (CMT or tungsten inert gas (TIG), for example), faster processing, and precise control of the process. RLW provides high flexibility in controlling the heat distribution (e.g., through beam shaping and beam

characterisations).

Table 3. List of parameters used for the welding trials.

	Zn coating	Oscillation freq. (Hz)	Welding speed (mm/s)	Focal offset (mm)	Oscillation amplitude (mm)	Laser power (W)
ID 1	Yes	100	30	-0.7	4	5250
ID 2	No	100				(core-to-ring ratio = 1.1)
ID 3	Yes	150				
ID 4	No	150				

Table 4. List of parameters for removing the Zn coating from the DC01 steel.

Oscillation Freq. (Hz)	Oscillation amplitude (mm)	Focal offset (mm)	Ablation speed (mm/s)	Laser power (W)
150	5	3	10	475 (core-to-ring ratio = ∞)

A high-speed camera Photron FASTCAM NOVA S6 coupled with CAVILUX Smart laser illumination (808 nm with exposure of 20 μ s) was adopted for the in-process monitoring of the welding process with a 7×9 mm² static field of view (FOV), centred at the middle of the weld. The camera's frame rate was 15,000 fps with a resolution of 768 × 592 pixels.

2.3. Weld characterisation and testing

Welded samples were subjected to microstructural observations, chemical characterisation and mechanical testing. The metallography process was implemented to obtain three cross-sections from each welded sample for microstructural observations and chemical composition analysis. This procedure started with cutting the samples from three surfaces (middle and 15 mm from each end) perpendicular to the welding direction. Specimens were mounted in a conductive Bakelite resin and ground using 400 and 2000 grit size SiC papers. Following 9 μ m and 3 μ m particles polishing, samples eventually were polished down to 0.02 μ m and no etching was used. A Keyence VHX7000 optical microscope was utilised for the topographic analysis of the weld bead (steel side) and the fracture surface (aluminium side), as well as weld profile. Furthermore, high magnification analysis using a JEOL 7800F scanning electron microscope (SEM) was used to reveal the IMCs. ImageJ software was employed for all the microstructural calculations, i.e., roughness features and IMCs thickness. The energy-dispersive X-ray spectroscopy (EDS) analysis was implemented for analysing the chemical composition of the weld. A 20 kV accelerating voltage, a 16 nA current and up to 300,000 cps input count plus a dwell time of 250 ms was used for Fe and Al elemental maps and IMC analysis. Lap shear tensile tests were performed on three replications of each sample using an Instron 30 kN tensile machine at a 2 mm/min extension rate. Pearson's correlation was obtained to analyse the correlation between the key outputs. ANOVA was also used to test the statistical significance of the results.

3. Results and discussion

3.1. Topographic analysis and mechanisms of spattering

Observation of the weld surface and interface can deliver information about the characteristics of the samples welded with and without Zn coating. Fig. 2 (a) shows the images of the top surface (steel-side) and interface (aluminium-side) after the tensile test of the welded samples. Weld instabilities are associated with higher values of the total humps and holes area.

The connection area at the aluminium-side in Fig. 2 (a) is enveloped by the black lines in the topography images. Fig. 3 shows the quantitative analysis of key welding features at the top surface (steel) and the interface (aluminium). The following observations are made:

- significant molten pool instabilities are seen in Zn-coated samples, especially when the welding frequency is 150 Hz (ID 3) - the total humps and holes area are higher in this sample;

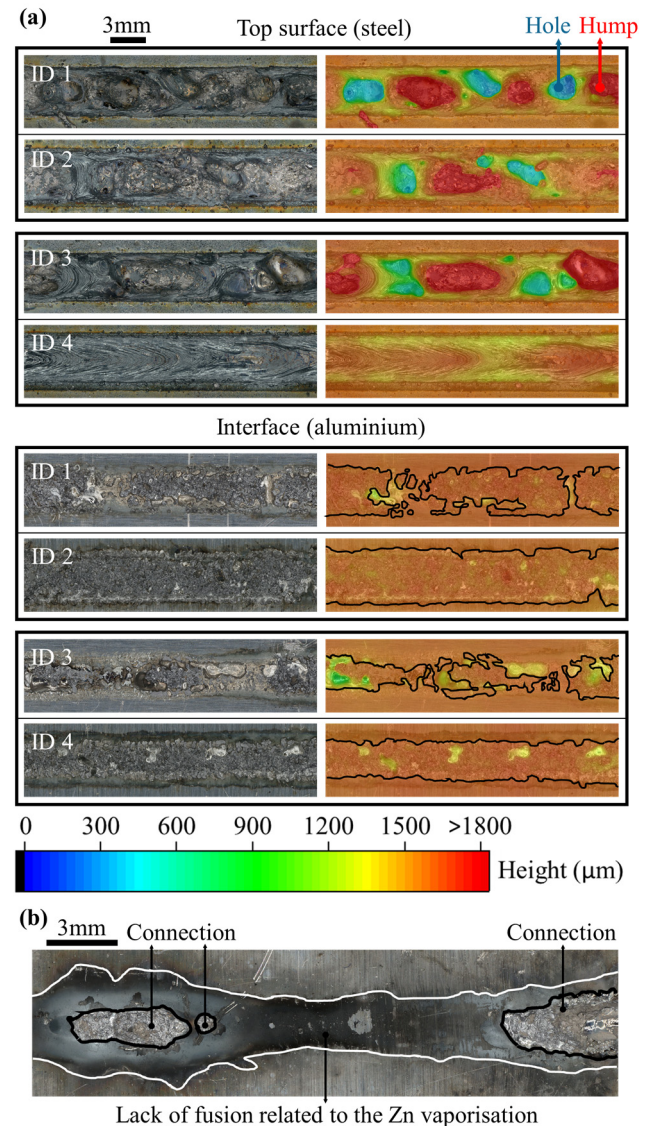


Fig. 2. (a) Topographic analysis of the top surface (steel-side) and the interface (aluminium-side) after tensile test. (b) Detailed view of the local explosions caused by the Zn vapours at the aluminium interface.

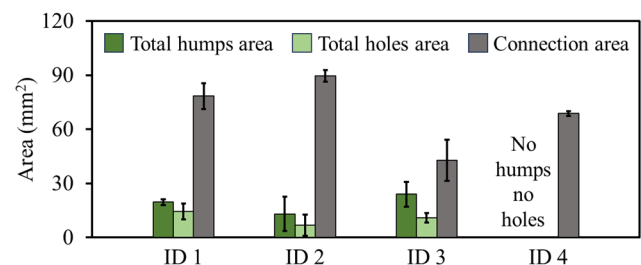


Fig. 3. Quantitative analysis of the key welding features at the top surface (steel) and the interface (aluminium).

- welding instabilities in samples without Zn (ID 2 and 4) are much lower compared to the ones with Zn coating. Indeed, no welding instabilities were observed when oscillation frequency increased from 100 Hz (ID 2) to 150 Hz (ID 4) - this trend is opposed to the one observed with Zn-coated samples (ID 1 and 3). At an oscillation frequency of 150 Hz, the distance between oscillation cycles (corresponding to the oscillation period) is reduced. This shorter distance causes the same area of the material to be reheated multiple times, allowing it to melt more evenly and thoroughly. As the laser moves away, the molten material flows back into the melt pool, filling any gaps or uneven areas - this is the reason for a smooth surface in ID 4 (no Zn coating). However, in the sample with Zn coating, higher oscillation frequency leads to an unstable surface due to more turbulent fluid flow. This can be due to the fact that the higher beam oscillation frequency combined with Zn vapour causes more agitation in the weld pool; and
- therefore, unlike ID 4, more humps and holes are seen in the Zn-coated sample (ID 3). This observation indicates increased irregularities in the Zn-coated sample. At each oscillation frequency, the connection area in the Zn-coated samples (ID 1 and 3) is smaller compared to the samples without Zn coating (ID 2 and 4). The role of the Zn vapour during the welding process leads to spattering and material ejection in the weld pool. Zn vapour interacts with the laser beam and causes beam absorption or deflection as well as local explosions [10], which prevents the melting of the aluminium sheet. This is visible by the black marks left at the aluminium interface in Fig. 2 (b). These local explosions cause un-even connection areas in the joints with Zn coating and highly influence the joint strength and repeatability.

To further discuss these findings, Fig. 4 demonstrates the high-speed images taken from the top surface (steel) of the weld pool in ID 1 and ID 2. Several features including, spattering, lack of fusion, and weld pool instabilities are recognisable from this comparison. Three mechanisms of spattering were distinguished during the welding process.

- Type I spattering* - initially, clusters of tiny spatters (shown in green-dashed lines) happen mostly when the beam proceeds and melts the material. This can be seen clearly in the sample without Zn due to its black background (because of the Zn removal process), but it does happen in the sample with Zn as well.
- Type II spattering* - the second type of spattering that causes the most severe material ejection, happens at the edges of the weld pool when the beam changes its direction and leaves that edge. As the laser beam moves toward the reversal points, liquid metal follows it to the edge zone of the molten pool, pulled by the inertia forces caused by the lateral acceleration. The intensive movement of the weld pool leads to material ejection and stops the liquid column from flowing back into the molten pool. This is also one main reason for the presence of surface instabilities leading to the lack of fusion. It should be noted that the applied heat input is also higher at the turning point caused by the reversal of the direction of the laser beam motion, and this exacerbates the spattering from edges [11].

- Type III spattering* - the third type of the spattering, which is observed only in the sample with Zn (ID 1) is the gas bubble entrapping the molten material or spattering because of the Zn vaporisation. One example of this type can be seen in steps 8 and 9 in Fig. 4 of the sample with Zn, where a large round gas bubble bursts into several smaller spatters.

Like type I, type II happens in all of the samples regardless of the presence or absence of Zn coating. However, this kind of spattering happens severely when the oscillation frequency is 100 Hz because of the higher heat input - this can justify the presence of the welding instabilities in ID 2. Furthermore, in Fig. 4, the lack of fusion and the presence of the holes are marked with blue line. These features happen due to the material's removal from the weld pool (spattering) and the molten material's failure to fill the backside area, as it was discussed earlier. The lack of fusion and the surface holes/humps happen both with and without Zn, but they are clearly more intense in the samples with Zn coating due to more agitation in the molten pool.

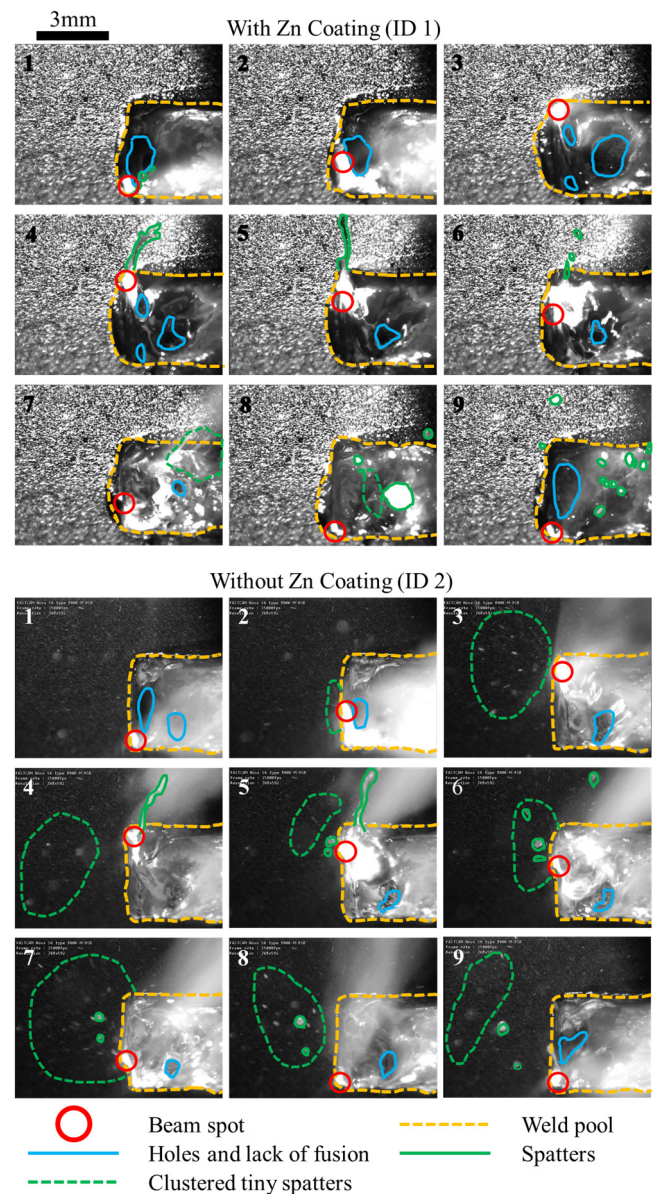


Fig. 4. High speed images for samples with Zn coating (ID 1) and without Zn coating (ID 2). The figure shows the phenomena in a 9 consecutive time steps and in a total time frame of 1 ms. High speed video is available from this [link](#).

3.2. Microstructural observations

The cross-sections of the joint in the samples with and without Zn coating are indicated in Fig. 5. Once again, it can be seen that Zn vaporisation results in a deficient weld pool with a lack of fusion in some regions and several pores. In the samples without Zn, higher seam quality and uniform connection are observed with a minimum number of pores or lack of fusion. These differences are ascribed to more agitation in the weld pool and an un-even connection area due to the Zn vapours. Besides, relatively more mixing happens in the samples with Zn coating, as seen in elemental maps in Fig. 5. Various types of the IMCs (continuous layers of Fe_2Al_5 , FeAl_2 and FeAl plus the needle-like $\text{Fe}_4\text{Al}_{13}$) can be identified in an area of the joint edge, where the maximum mixing happens.

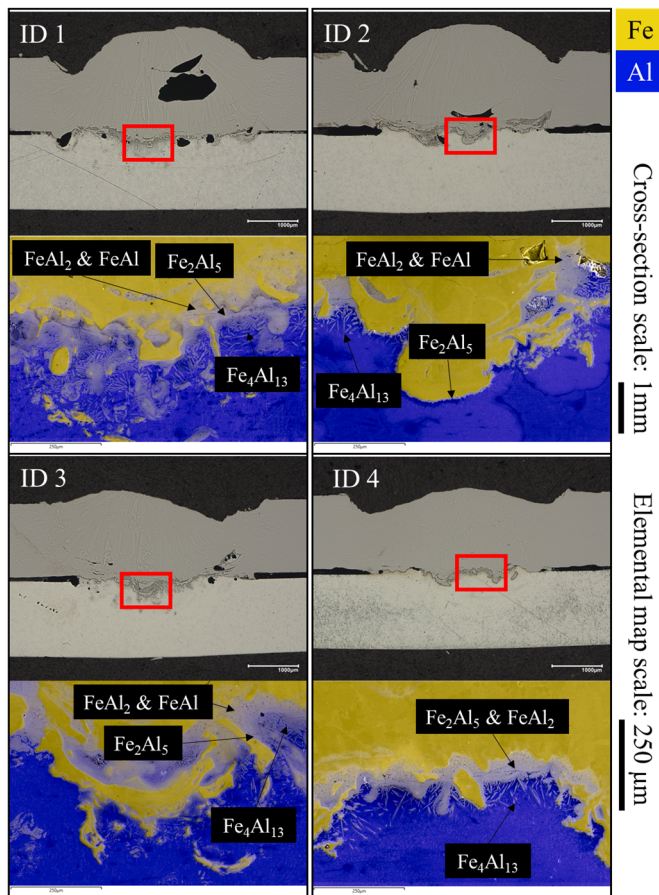


Fig. 5. Cross-section of the samples with (ID 1 and 3) and without Zn coating (ID 2 and 4), and the elemental map analysis in the area depicted in red.

It is worth noting that due to the non-uniform connection in the samples with Zn coating (ID 1 and ID 3), regions with minimum mixing are expected due to the lack of or reduced connection. However, when maximum mixing is considered, this also happens in the samples with Zn coating as a deeper penetration depth is reached in these samples. Mixing leads to a higher interaction of Fe and Al atoms and formation of a thicker IMC layer. As the thickness of IMCs increases, the susceptibility of the cracking grows as well [12]. The thickness of the total IMCs and the Fe_2Al_5 phase (the most brittle IMC phase [13]) are reported in Fig. 6. The higher mixing of the Fe and Al in the samples with Zn coating can be witnessed in Fig. 6 as higher total IMCs thickness was detected in these samples compared to the samples without Zn coating. Zn-

coated specimens (ID 1 and 3) also exhibit more variability in IMCs thickness, which can be attributed to the different penetration depths and mixing levels in Zn-coated specimens. Unlike the total IMCs thickness, the thickness of the Fe_2Al_5 phase is in a similar range for the four samples.

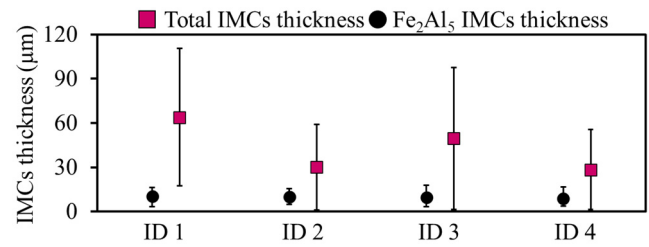


Fig. 6. Thickness of the total IMC and Fe_2Al_5 phase.

3.3. Mechanical properties

The lap shear tensile results for the welded samples are reported in Fig. 7. At first sight, it is clear that the average linear load is higher for the samples without Zn coating. Furthermore, the spread in the linear load for the specimens without Zn coating is less, which translates to higher repeatability. The main reason for this can be explained by considering the topographic and microstructural properties of the samples (Fig. 3 and 5). According to the topography information from the interface (aluminium), less connection area is present in Zn-coated samples (ID 1 and 3). Therefore, the joint is weaker compared to the samples without Zn. This is considered the primary cause of lower strength in the samples with Zn coating.

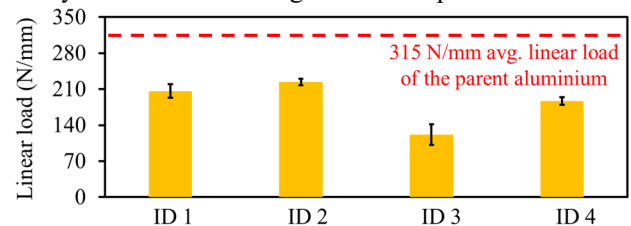


Fig. 7. Lap shear tensile test results for the samples with (ID 1 and 3) and without Zn coating (ID 2 and 4).

The large spread in the joint strength in Zn-coated samples (ID 1 and 3) can be justified by the unstable molten pool behaviour as already observed in previous sections, leading to an un-even connection area and ultimately severe and localised mixing as shown by the total IMCs thickness in Fig. 6. It can be seen that the linear load is higher when the oscillation frequency is 100 Hz in both conditions with and without Zn. This is because of the large connection area in these samples, unlike the shallow or inadequate penetration that happens using higher oscillation frequency (Fig. 3 and 5). On average, the highest linear load of 225 N/mm was achieved in the sample without Zn coating in ID 2 (100 Hz), which yields 72% of joint efficiency compared to the weakest parent material (5251-H22 aluminium with average 315 N/mm UTS as in Table 1). The highest linear load obtained in the Zn-coated sample was 206 N/mm for ID 1 (100 Hz) – this corresponds to an improvement of joint strength of approx. 10% when comparing ID 1 to ID 2. The strength improvement is more significant when comparing ID 3 and ID 4 (150 Hz) with approx. 50%.

The correlation between the joint strength and the connection area is 0.99 (Fig. 8 (a)), representing a clear link between them. Also, the correlation between joint strength and

weld depth and Fe_2Al_5 thickness are -0.16 (Fig. 8 (b)) and 0.14 (Fig. 8 (c)), respectively. However, no evident correlation was found between joint strength and total IMC thickness. Joint strength is slightly correlated with welding depth and Fe_2Al_5 thickness. Although a negative correlation with welding depth was expected, it is negligible, possibly due to the greater influence of the connection area on the strength. Besides, the narrow range of Fe_2Al_5 thickness (consistently below 10 μm) likely explains its small correlation with joint strength; research in [14] highlighted that a significant drop in joint strength is only visible when the thickness of Fe_2Al_5 exceeds 10 μm .

To further analyse the effect of Zn coating and oscillation frequency on the responses (joint strength, welding depth, total IMCs thickness and Fe_2Al_5 thickness), an ANOVA test was run. Results are shown in Table 5, testing the hypothesis that the samples within the setup are having the same mean against the alternative hypothesis that means are not all the same; if the p-value is higher than 0.1, the hypothesis is accepted. According to Table 5, the presence or absence of a Zn coating and the utilisation of the two different oscillation frequencies of 100 and 150 Hz have a significant influence on the linear load, total IMCs thickness and Fe_2Al_5 IMCs thickness. Accordingly, welding depth is not affected by the oscillation frequency (p-value = 0.5658), although it is affected by the Zn coating.

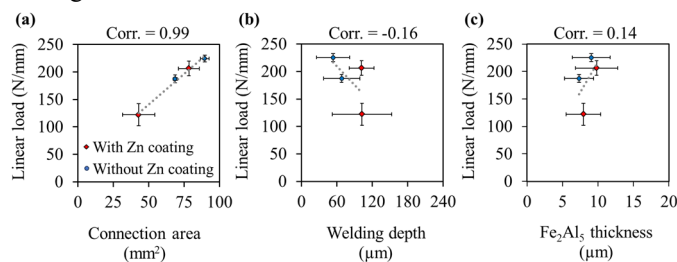


Fig. 8. Correlation of linear load with (a) connection area, (b) average total IMCs thickness and (c) average Fe_2Al_5 thickness.

Table 5. Summary of the ANOVA tests with p-values.

Responses	Source		
	Zn coating	Oscillation freq.	Interaction
Linear load	0.0005	0.0	0.0161
Welding depth	0.0038	0.5658	0.594
Total IMCs thickness	0.0	0.0145	0.0595
Fe_2Al_5 IMCs thickness	0.0877	0.0	0.8564

4. Conclusions

This paper studied the impact of Zn coating during RLW of Zn-coated steel to aluminium in a nominal zero part-to-part gap lap configuration. Zn coating was removed in the first pass in half of the samples to compare the influence of the Zn on the weld strength and the process repeatability. Welds were produced in the second pass. Key findings are summarised as follows:

- the presence of Zn coating results in an un-even connection area between steel and aluminium. This is explained by the localised explosions, which prevents the melting of the aluminium sheet. As a result, the connection area is significantly lower when Zn coating is present;
- three types of spattering have been observed and they are associated with the combined effect of the localised pressure exerted by the Zn vapour and the inertia forces

induced by the local variations of the laser beam velocity across the oscillation path;

- Zn vapours cause instability to the molten pool leading to an increase in average and variation of the IMCs thickness; and,
- significant joint efficiency up to 72% is achieved when the Zn coating is removed, opposed to 65% when the Zn is present. Joint strength is predominantly correlated with the connection area (0.99 Pearson's correlation).

Future work will focus on enlarging the window of process parameters (lower than 100 Hz and higher than 150 Hz beam oscillation frequency) and test the response of the welded samples to corrosion protection when the Zn coating is locally removed to improve weldability. Additionally, future endeavours will be devoted to study the complex fluid-flow mechanism between steel and aluminium by employing laser beam oscillation and beam shaping (both static and dynamic).

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